

SYDNEY TECHNICAL HIGH SCHOOL



TRIAL HIGHER SCHOOL CERTIFICATE

2003

MATHEMATICS EXTENSION 2

General Instructions:

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Approved calculators may be used
- All necessary working should be shown in every question
- A table of standard integrals is supplied at the back of this paper
- Start each question on a new page
- Attempt all Questions 1 – 8
- All questions are of equal value
- **Total marks 120**

Name: _____

Class: _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Question 7	Question 8	TOTAL

Question 1

Marks

- a) (i) Find $\int \frac{1}{\cos x + 2} dx$ using the substitution $t = \tan \frac{x}{2}$ 3

Evaluate:

(ii) $\int_2^4 \frac{dx}{x^2 - 4x + 8}$ 3

(iii) $\int_{-1}^1 \frac{4+x^2}{4-x^2} dx$ 4

- b) Let n be a positive integer and let

$$I_n = \int_1^2 (\log_e x)^n dx$$

- (i) Prove that $I_n = 2(\log_e 2)^n - nI_{n-1}$ 2

- (ii) Hence evaluate $\int_1^2 (\log_e x)^4 dx$ as a polynomial in terms of $\log_e 2$ 3

Question 2

- a) The complex number z and its conjugate $\bar{z}$ satisfy the equation $z\bar{z} + 2iz = 12 + 6i$. Find the possible values of z . 4

- b) On an Argand diagram shade the region containing all points representing complex numbers z such that $\operatorname{Re}(z) \leq 1$ and $0 \leq \arg(z+i) \leq \frac{\pi}{4}$ 3

- c) Express $z_1 = \frac{7+4i}{3-2i}$ in the form $a+ib$ where a and b are real. 1

- d) On an Argand diagram sketch the locus of the point representing the complex number z such that $|z-3-i| = \sqrt{10}$. Find the greatest value of $|z|$ subject to this condition. 3

- e) (i) Given that w is a complex root of the equation $x^3 = 1$, show that w^2 is also a root of this equation. 2

- (ii) Show that $1+w+w^2=0$, and $1+w^2+w^4=0$. 2

Question 3

Marks

The ellipse E has Cartesian equation $\frac{x^2}{25} + \frac{y^2}{16} = 1$

a) Find

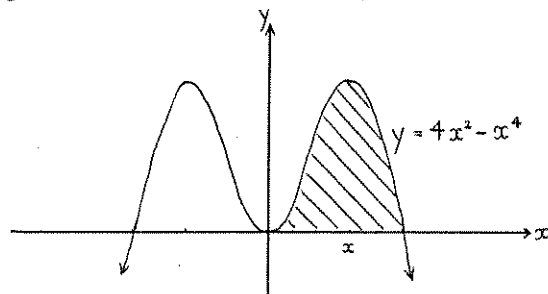
(i) the coordinates of the foci S and S' 1

(ii) Show that any point P on E can be represented by the coordinates $(5 \cos \theta, 4 \sin \theta)$ and hence or otherwise prove that $PS + PS'$ is a constant. 3

(iii) Show that the equation of the normal at the point P on the ellipse is $\frac{5x}{\cos \theta} - \frac{4y}{\sin \theta} = 9$ 3

(iv) If this normal meets the x axis at M and the y axis at N , prove that $\frac{PM}{PN} = \frac{16}{25}$ 4

b) The region shaded below is rotated about the y -axis to form a solid of revolution.



Using the method of cylindrical shells to calculate the volume of this solid, show that:

(i) The volume δV of a shell at x is given by $\delta V = 2\pi(4x^3 - x^5)\delta x$ 2

(ii) Hence find the volume of this solid. 2

Question 4

a) Let $f(x) = -x^2 + 8x - 12$. On separate diagrams, and without calculus, sketch the following graphs. Indicate clearly any asymptotes and intercepts with the axes.

(i) $y = f(x)$ 2

(ii) $y = |f(x)|$ 2

(iii) $y^2 = f(x)$ 2

(iv) $y = \frac{1}{f(x)}$ 2

(v) $y = e^{f(x)}$, giving the coordinates of any turning points by not using calculus. 3

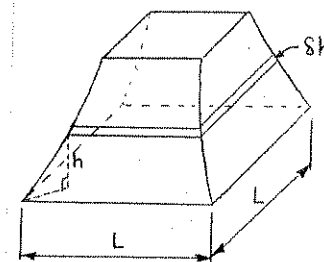
b) Given $p + q \geq 2\sqrt{pq}$ if p and q are positive real numbers

(i) Show that $e^a + e^b \geq 2e^{\frac{a+b}{2}}$ for all real a and b 2

(ii) Hence find the minimum value of $e^{-2x} + e^{-x} + 1 + e^x + e^{2x}$ for all real x . 2

Question 5

a)



A stone building of height H metres has the shape of a flat topped square 'pyramid' with curved sides as shown in the figure. The cross-section at height h metres is a square with sides parallel to the sides of the base and of length l , $l = \frac{L}{\sqrt{h+1}}$ where L is the side length of the square base in metres.

(i) Write an expression for the volume of a slice at height h metres. 2

(ii) Hence find the volume of the building in terms of L and H . 2

Question 5

b) The Fibonacci Sequence, F_n , is defined by:

$$F_1 = 1$$

$$F_2 = 1$$

$$F_{n+2} = F_{n+1} + F_n \text{ for all } n \geq 1$$

(i) Write down the first 12 terms of the sequence

1

(ii) Prove, by mathematical induction, that for all positive integers, n , F_{4n} is divisible by 3.

5

c) Find $\int \frac{\sin^3 \theta}{\cos^4 \theta} d\theta$

3

d) Consider the function of $y = \tan^{-1}(\tan x)$

(i) What is its period?

1

(ii) Hence sketch the function for $-2\pi \leq x \leq 2\pi$

1

Question 6

The equation $x^3 + 2x - 1 = 0$ has roots α , β , and γ . In each of the following cases, find an equation with integer coefficients having the roots stated below.

a) (i) $-\alpha, -\beta, -\gamma$

1

(ii) $\alpha, -\alpha, \beta, -\beta, \gamma, -\gamma$

2

(iii) $\alpha^2, \beta^2, \gamma^2$

3

b) (i) Prove that 1 and -1 are both roots of multiplicity 2 of the polynomial $P(x) = x^6 - 3x^2 + 2$

2

(ii) Express $P(x)$ as the product of irreducible factors over the field of

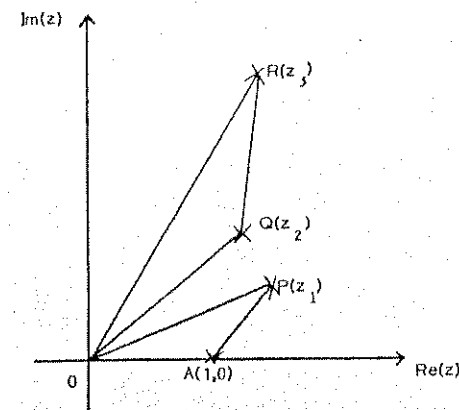
(α) rational numbers

1

(β) complex numbers

1

c)



In the Argand diagram above, ΔOQR is constructed similar to ΔOAP .

Show that

(i) $|z_3| = |z_1| |z_2|$

2

(ii) $\arg z_3 = \arg z_1 + \arg z_2$

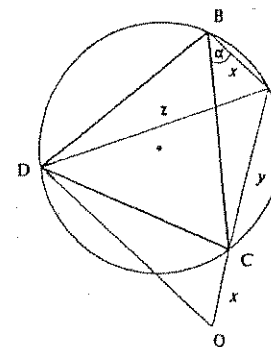
2

(iii) What is the significance of these results?

1

Question 7

The figure shows two towns located at B and C. BCD is an equilateral triangle. A road junction is to be placed at P, somewhere on the minor arc BC of the circumscribed circle of the triangle BCD.



Let BP, CP and DP have lengths x , y , z respectively.

The point Q is on the line PC, extended so that BP and CQ have the same length x . Let $\angle PBC = \alpha$.

Question 7 (cont)

- a) (i) Show that $\angle BPD = \angle CPD = 60^\circ$ 2
- (ii) Find $\angle DCQ$ in terms of α 1
- (iii) Prove $\triangle PBD \cong \triangle QCD$. 2
- (iv) Prove $\triangle DPQ$ is equilateral 2
- (v) Now show that $z = x + y$ 1
- b) Owing to the tides, the depth of water in an estuary may be assumed to rise and fall with time in simple harmonic motion.
- At a certain place there is a danger of flooding when the depth of the water is above 1.25m. One day high tide was 1.5m at 1am and the following low tide was 0.5m at 7:30am.
- (i) Find the amplitude in metres and period in minutes of this tidal motion. 2
- (ii) Hence find between what times after 1am was there no danger of flooding. 3
- c) Find $\int \frac{1-x}{1-\sqrt{x}} dx$ 2

Question 8

- a) (i) Find the 1st and 2nd derivatives of $P(x) = \frac{x^4}{4} + \frac{x^3}{3} + \frac{x^2}{2} + x + c$ 1
- (ii) Hence or otherwise show that $P(x) = 0$ has not real roots if $c > \frac{7}{12}$ 3
- b) (i) Write down in mod-arg form, the five roots of $z^5 - 1 = 0$ 3
- (ii) By combining appropriate pairs of these roots, show that for $z \neq 1$,

$$\frac{z^5 - 1}{z - 1} = (z^2 - 2z \cos \frac{2\pi}{5} + 1)(z^2 - 2z \cos \frac{4\pi}{5} + 1)$$
 4
- (iii) Deduce that $\cos \frac{2\pi}{5}$ and $\cos \frac{4\pi}{5}$ are the roots of the equation 4
- $$4x^2 + 2x - 1 = 0$$

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Solutions to 2003 S.T.H.S. 4 Trial

i. a. (i) $\int \frac{1}{\cos x + 2} dx$

$$= \int \frac{1}{\frac{1-t^2}{1+t^2} + 2} \times \frac{2dt}{1+t^2} \quad (1)$$

$$= \int \frac{2dt}{1-t^2 + 2(1+t^2)}$$

$$= \int \frac{2dt}{3+t^2} \quad (1)$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \frac{t}{\sqrt{3}} + C$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{\tan \frac{x}{2}}{\sqrt{3}} \right) + C \quad (1)$$

ciii) $\frac{4+x^2}{4-x^2} = \frac{8-(4-x^2)}{4-x^2} = \frac{8}{4-x^2} - 1$

Let $\frac{8}{4-x^2} = \frac{A}{2-x} + \frac{B}{2+x}$

Then $8 = A(2+x) + B(2-x)$

Let $x = -2, \therefore B = 2 \quad (1)$

$x = 2, \therefore A = 2 \quad (1)$

cii) $\int_2^4 \frac{dx}{x^2 - 4x + 8}$

$$\int_2^4 \frac{dx}{(x-2)^2 + 4} \quad (1)$$

$$\left[\frac{1}{2} \tan^{-1} \frac{x-2}{2} \right]_2^4 \quad (1)$$

$$= \frac{1}{2} \tan^{-1} 1 - \frac{1}{2} \tan^{-1} 0$$

$$= \frac{\pi}{8} \quad (1)$$

$\therefore \int_{-1}^1 \frac{1}{x^2} dx$ becomes

$$\int_{-1}^1 \frac{2}{2-x} + \frac{2}{2+x} - 1 dx$$

$$= \left[-2 \log_e (2-x) + 2 \log_e (2+x) - x \right]_{-1}^1 \quad (1)$$

$$= -2 \log_e 1 + 2 \log_e 3 - 1 - (-2 \log_e 3 + 2 \log_e 1 + 1)$$

$$= 4 \log_e 3 - 2 \quad (1)$$

cii) a) $I_n = \int_1^2 (\log_e x)^n dx$

$$= \int_1^2 (\log_e x)^n \frac{d}{dx} x dx$$

$$= \left[x (\log_e x)^n \right]_1^2 - \int_1^2 x \cdot n (\log_e x)^{n-1} \cdot \frac{1}{x} dx \quad (1)$$

$$= 2 (\log_e 2)^n - n I_{n-1} \quad \text{by parts} \quad (1)$$

b) $\int_1^2 (\log_e x)^4 dx = I_4$

$$= 2 (\log_e 2)^4 - 4 I_3$$

$$= 2 (\log_e 2)^4 - 4 \left[2 (\log_e 2)^3 - 3 I_2 \right]$$

$$= 2 (\log_e 2)^4 - 8 (\log_e 2)^3 +$$

$$12 \left[2 (\log_e 2)^2 - 2 I_1 \right] \quad (1)$$

Now $I_1 = \int_1^2 \log_e x \, dx$
 $= [x \log_e x - x]_1^2$ by parts ①
 $= 2 \log_e 2 - 1$

$\int_1^2 (\log_e x)^4 \, dx = \underline{2(\log_e 2)^4 - 8(\log_e 2)^3}$ ①
 $\underline{+ 24(\log_e 2)^2 - 48 \log_e 2 + 24}$

2. (a) Let $z = x + iy$
 $\therefore z\bar{z} + 2iz = 12 + 6i$ becomes

$(x+iy)(x-iy) + 2i(x+iy) = 12 + 6i$

$x^2 + y^2 + 2ix - 2y = 12 + 6i$

$\therefore x^2 + y^2 - 2y = 12$ and $2x = 6$ ①

$\therefore x^2 + y^2 - 2y = 12$ and $x = 3$ ①

$y^2 - 2y - 3 = 0$

$(y+1)(y-3) = 0$

$y = -1$ or 3

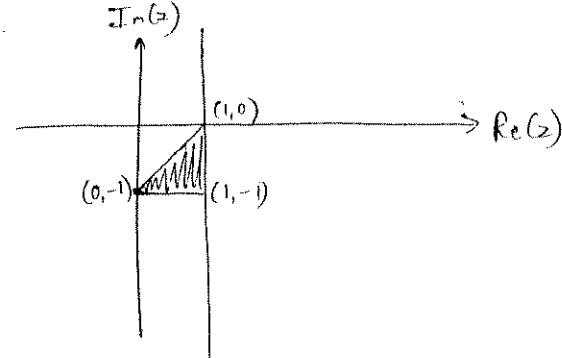
$\therefore z = 3 - i$ or $3 + 3i$ ②

3

(b) $\operatorname{Re}(z)$ ①

$0 \leq \arg(z+i) \leq \frac{\pi}{4}$

$0 \leq \arg(z - (0-i)) \leq \frac{\pi}{4}$ ①



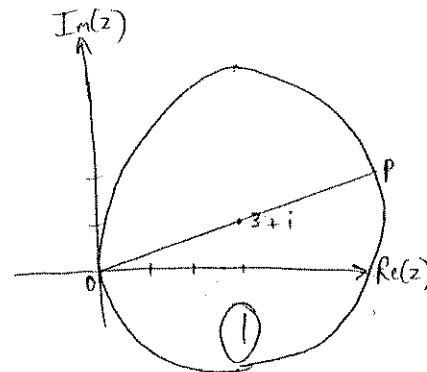
②

(c) $z_1 = \frac{7+4i}{3-2i} \times \frac{3+2i}{3+2i}$ ①
 $= \frac{13+26i}{13} = 1+2i$

(d) $|z - 3 - i| = \sqrt{10}$

$\therefore |z - (3+i)| = \sqrt{10}$

is a circle centre $(3, 1)$ radius $\sqrt{10}$ ①



$(0, 0)$ lies on the circle. Greatest value of $|z| = OP$

$= 2\sqrt{10}$ ①

Q (i) If ω is a root of $x^3=1$ then ω satisfies the equation

$$\text{ie: } \omega^3 = 1$$

$$\therefore (\omega^3)^2 = \omega^6 = 1 \quad \text{ie: } (\omega^2)^3 = 1$$

ie: ω^2 also satisfies $x^3=1$.

(ii) The sum of the roots of

$$x^3 - 1 = 0 \quad \text{is } -\frac{b}{a} \quad \text{ie: } 0$$

$$\therefore \underline{1 + \omega + \omega^2 = 0.}$$

$$\text{Since } \omega^3 = 1, \\ \omega^4 = \omega$$

$$\therefore \underline{1 + \omega^4 + \omega^2 = 0.}$$

3. (i) $a=5$ $b=4$

$$e^2 = 1 - \frac{b^2}{a^2}$$

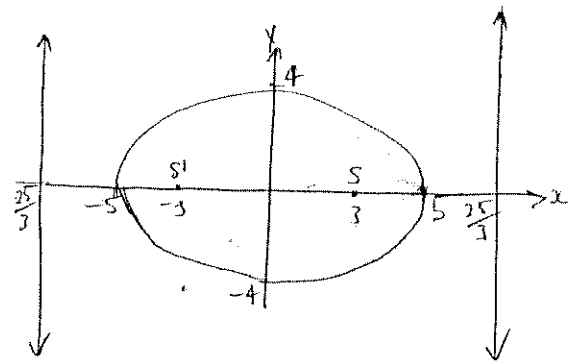
$$= 1 - \frac{16}{25}$$

$$e^2 = \frac{9}{25}$$

$$e = \frac{3}{5}$$

$$\text{Q (ii) } S(ae, 0) \quad S'(-ae, 0)$$

$$\underline{S(3, 0) \quad S'(-3, 0)} \quad (1)$$



(ii) Parametric Form of an ellipse

$$(a \cos \theta, b \sin \theta)$$

$$a=5, \quad b=4$$

$$\therefore \underline{P(5 \cos \theta, 4 \sin \theta)} \quad (1)$$

$$PS + PS'$$

$$eP_{\text{directrix}_1} + eP_{\text{directrix}_2}$$

$$e\left(\frac{25}{3} - 5\cos\theta\right) + e\left(5\cos\theta + \frac{25}{3}\right) \quad \text{from the def'n of an ellipse}$$

$$= 2e \times \frac{25}{3}$$

$$= 2 \times \frac{4}{5} \times \frac{25}{3}$$

$$= 10 \quad (1)$$

$$\text{iii)} \quad \frac{x^2}{25} + \frac{y^2}{16} = 1$$

Differentiating implicitly y ,

$$\frac{2x}{25} + \frac{2y \frac{dy}{dx}}{16} = 0$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{-2x}{25} \times \frac{16}{2y} \\ &= \frac{-16x}{25y} \end{aligned}$$

$$\text{at } P, \quad \frac{dy}{dx} = M_{\text{tangent}}$$

$$= \frac{-16 \times 5\cos\theta}{25 \times 4\sin\theta}$$

$$= \frac{-4\cos\theta}{5\sin\theta}$$

$$M_{\text{tangent}} = -\frac{4}{5\tan\theta} \quad (1)$$

Eq'n of normal:

$$y - 4\sin\theta = \frac{5\sin\theta}{4\cos\theta}(x - 5\cos\theta) \quad (1)$$

$$4y\cos\theta - 16\sin\theta\cos\theta = 5\sin\theta x - 25\sin\theta\cos\theta$$

$$4y\cos\theta = 5\sin\theta x - 9\sin\theta\cos\theta$$

$$4y = \frac{5\sin\theta x - 9\sin\theta\cos\theta}{\cos\theta}$$

$$y = \frac{5x}{4\cos\theta} - \frac{9\sin\theta}{4} \quad (1)$$

iv) Cuts x axis when $y=0$

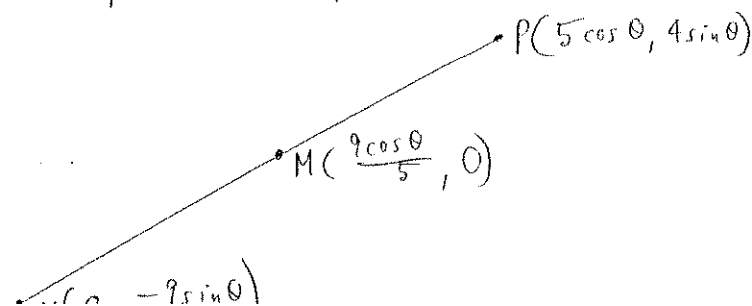
$$\text{ie: } 0 = \frac{5x}{4\cos\theta}$$

$$x = \frac{9\cos\theta}{5} \quad (1)$$

Cuts y axis when $x=0$

$$0 = -\frac{9\sin\theta}{4}$$

$$y = -\frac{9\sin\theta}{4} \quad (1)$$



$$\frac{PM}{PN}$$

$$= \frac{\sqrt{(5\cos\theta - \frac{2}{5}\cos\theta)^2 + (4\sin\theta)^2}}{\sqrt{(5\cos\theta)^2 + (4\sin\theta + (\frac{9\sin\theta}{4}))^2}}$$

$$= \frac{\sqrt{(\frac{16}{5}\cos\theta)^2 + 16\sin^2\theta}}{\sqrt{25\cos^2\theta + (\frac{25}{4}\sin\theta)^2}}$$

$$= \frac{\sqrt{\frac{256}{25}\cos^2\theta + 16(1-\cos^2\theta)}}{\sqrt{25\cos^2\theta + \frac{625}{16}(1-\cos^2\theta)}}$$

$$= \sqrt{\frac{-5\frac{16}{25}\cos^2\theta + 16}{-14\frac{1}{16}\cos^2\theta + \frac{625}{16}}} \times \frac{400}{400} \quad (1)$$

$$= \sqrt{\frac{-2304\cos^2\theta + 6400}{-5625\cos^2\theta + 15625}}$$

$$= \sqrt{\frac{256(25 - 9\cos^2\theta)}{625(25 - 9\cos^2\theta)}}$$

$$= \frac{16}{25} \text{ as req'd.} \quad (1)$$

$$\therefore \text{ii) } \delta V = \pi((x+\delta x)^2 - x^2) \cdot h \quad (1)$$

$$= \pi(x^2 + 2x\delta x + \delta x^2 - x^2)(4x^2 - x^4)$$

$$= 2\pi(4x^3 - x^5)\delta x \text{ since } \delta x^2 \rightarrow 0. \quad (1)$$

$$\text{ii) } V = \int_0^a 2\pi(4x^3 - x^5)dx \text{ where } a \text{ is}$$

where the curve cuts the x-axis

$$\text{ie: } 4x^2 - x^4 = 0$$

$$x^2(4 - x^2) = 0$$

$$x = 2$$

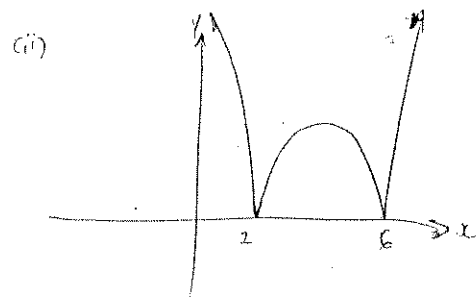
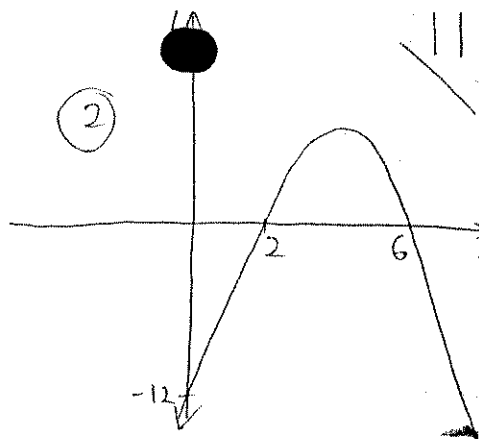
$$\therefore V = 2\pi \int_0^2 4x^3 - x^5 dx \quad (1)$$

$$= 2\pi \left[x^4 - \frac{x^6}{6} \right]_0^2$$

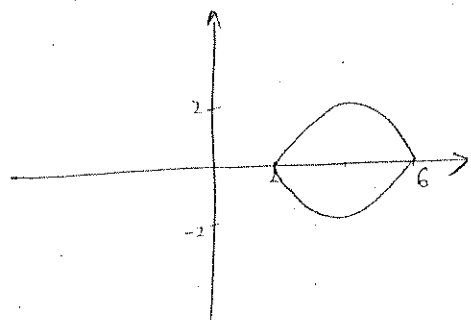
$$= 2\pi \left[16 - \frac{64}{6} \right]$$

$$= \frac{32\pi}{3} \text{ units}^3 \quad (1)$$

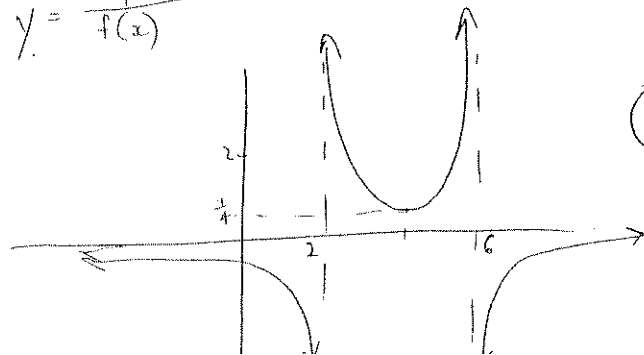
10 (i) $f(x) = -x^2 + 8x - 12$
 $= -(x^2 - 8x + 12)$
 $= -(x-2)(x-6)$



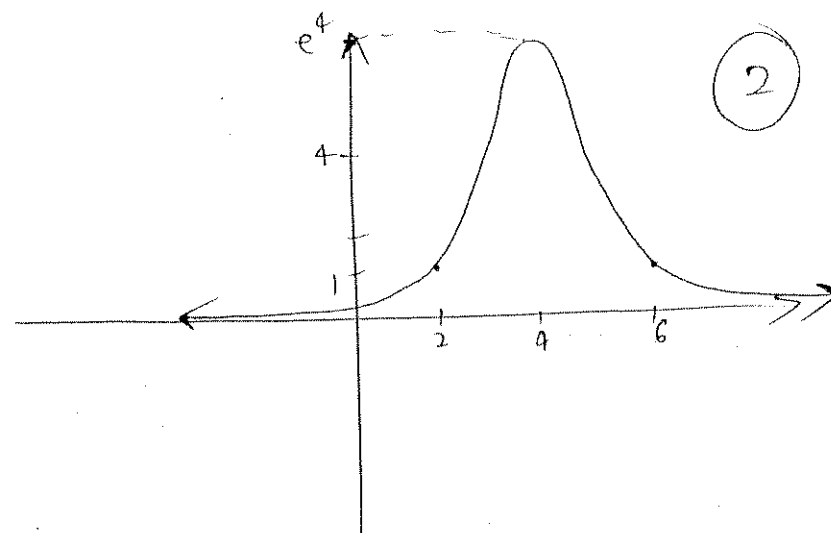
(iii) $y^2 = f(x)$
 $\Rightarrow y = \pm \sqrt{-x^2 + 8x - 12}$



(iv) $y = \frac{1}{f(x)}$



(v) $y = e^{-x^2 + 8x - 12}$



11 (i) If $p = e^a$, $q = e^b$, both real
 $e^a + e^b \geq 2(e^a \times e^b)^{\frac{1}{2}}$ from (i) ①
 $e^a + e^b \geq 2e^{\frac{a+b}{2}}$ ①

$$ii) e^{-2x} + e^{-x} + 1 + e^x + e^{2x} \geq 2e^{-\frac{x+x}{2}} + 1 + 2e^{\frac{-x+2x}{2}} \quad (1)$$

$$\geq 2 + 1 + 2$$

$$\geq 5$$

$\therefore$ Min. value is 5 (1)

5. Consider a slice, of thickness δh , at height h .

$$\text{Area of cross-section} = L^2 = \frac{L^2}{h+1} \text{ m}^2 \quad (1)$$

$$\therefore \text{Volume of slice} = \frac{L^2}{h+1} \delta h \text{ m}^3 \quad (1)$$

$$\begin{aligned} \therefore \text{Volume of solid} &= \int_0^H \frac{L^2}{h+1} dh \quad (1) \\ &= [L^2 \log_e(h+1)]_0^H \\ &= \underline{L^2 \log_e(H+1)} \quad (1) \end{aligned}$$

$$(b) (i) 1, 1, 2, 2, 5, 8, 13, 21, 34, 55, 89, 144 \quad (1)$$

ii) Show true for $n=1$.

$$F_4 = 3$$

$$3 = 3 \quad \checkmark$$

(1)

Assume true for $n=k$

ie: $F_{4k} = 3K$ where K is a positive integer (1)

Need now to show result holds for $n=k$.

ie: $F_{4(k+1)} = 3L$ where L is a positive integer.

$$\begin{aligned} \text{LHS } F_{4k+4} &= F_{4k+3} + F_{4k+2} \text{ from def'n. (1)} \\ &= F_{4k+2} + F_{4k+1} + F_{4k+1} + F_{4k} \\ &= F_{4k+1} + F_{4k} + F_{4k+1} + F_{4k+1} + F_{4k} \\ &= 3F_{4k+1} + 2F_{4k} \quad (1) \\ &= 3F_{4k+1} + 6K \text{ (from assumption)} \\ &= 3[F_{4k+1} + 2K] \\ &= 3L \text{ as } F_{4k+1} + 2K \text{ is integer} \quad (1) \end{aligned}$$

∴ Since result is true for $n=1$, $\sqrt{15}$
 it must also be true for $n=1+1=2$,
 $n=2+1=3$ etc. for all positive integral
 values of n .

$$\textcircled{3} \int \frac{\sin^3 \theta}{\cos^4 \theta} d\theta$$

$$= \int \frac{\sin^2 \theta}{\cos^4 \theta} \cdot \sin \theta d\theta$$

$$= \int \frac{1 - \cos^2 \theta}{\cos^4 \theta} \sin \theta d\theta \quad \textcircled{1}$$

$$\text{Let } u = \cos \theta \quad \therefore du = -\sin \theta d\theta$$

$$= \int \frac{1 - u^2}{u^4} \cdot -du \quad \textcircled{1}$$

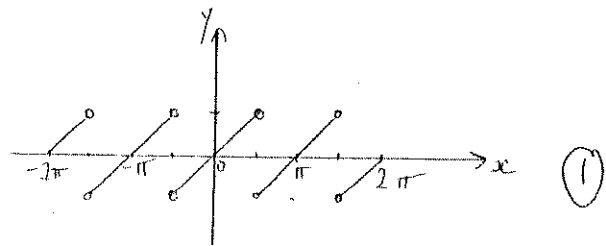
$$= \int \frac{1}{u^2} - \frac{1}{u^4} du$$

$$= -\frac{1}{u} + \frac{1}{3u^3} + C$$

$$= \underline{\underline{\frac{1}{3} \sec^3 \theta - \sec \theta + C}} \quad \textcircled{1}$$

∴ Period is π (same as for $\tan x$) $\textcircled{1}$

(ii)



6. (a) (i) $x^3 + 2x - 1 = 0$ has roots
 α, β, γ .

$$\therefore (-x)^3 + 2(-x) - 1 = 0$$

$$\text{ie: } \underline{x^3 + 2x + 1 = 0} \quad \textcircled{1}$$

(ii) $(x^3 + 2x + 1)(x^3 + 2x - 1) = 0$ from given
 equation and (i) $\textcircled{1}$

$$\text{ie: } \underline{x^6 + 4x^4 + 4x^2 - 1 = 0} \quad \textcircled{1}$$

$$\text{(iii)} \quad (x^{\frac{1}{2}})^3 + 2(x^{\frac{1}{2}}) - 1 = 0$$

$$x^{\frac{3}{2}} + 2x^{\frac{1}{2}} = 1$$

$$x^{\frac{1}{2}}(x + 2) = 1 \quad \textcircled{1}$$

$$x(x + 2)^2 = 1 \quad \textcircled{1}$$

$$x(x^2 + 4x + 4) = 1$$

$$\underline{x^3 + 4x^2 + 4x - 1 = 0} \quad \textcircled{1}$$

$$\textcircled{b} \text{ (i)} P(x) = x^6 - 3x^2 + 2$$

$$P'(x) = 6x^5 - 6x$$

$$= 6x(x^4 - 1)$$

$$= 6x(x^2 - 1)(x^2 + 1)$$

Since $(x-1)$, $(x+1)$ are factors of $P'(x)$ then $x = \pm 1$ are roots of multiplicity 2. ①

$$\Rightarrow P(x) = (x-1)^2(x+1)^2(x^2+2)$$

$$\text{a) } P(x) = (x-1)(x+1)(x-1)(x+1)(x^2+2) \quad ①$$

$$\text{b) } P(x) = (x-1)^2(x+1)^2(x+2i)(x-2i) \quad ①$$

c) i) By similar Δ 's ①

$$\frac{|z_3|}{|z_1|} = \frac{|z_2|}{1}$$

$$\therefore |z_3| = |z_1||z_2| \quad ①$$

$$\text{ii) } \angle ROQ = \angle POA \quad (\text{corr. } \angle\text{'s in similar } \Delta\text{'s}) \quad ①$$

$$\therefore \arg z_3 - \arg z_2 = \arg z_1$$

$$\therefore \arg z_3 = \arg z_1 + \arg z_2 \quad ①$$

iii) The construction can be used to multiply complex numbers ①

$$\text{i) } \angle CPD = \angle CBD \quad (\text{Angles in same segment}) \quad \therefore = 60^\circ \quad (\text{as } \Delta CBD \text{ is equilateral}) \quad ①$$

$$\angle BPD = \angle BCD \quad (\text{Angles in same segment}) = 60^\circ \quad (\text{as } \Delta CBD \text{ is equilateral}) \quad ①$$

$$\text{ii) } \angle BCQ = \angle CBP + \angle BPC \quad (\text{exterior angle})$$

$$\angle DCQ + 60 = \alpha + 120 \quad \text{from (i)}$$

$$\therefore \angle DCQ = \alpha + 60 \quad ①$$

$$\text{iii) } BD = DC \quad (\text{equilateral triangle})$$

$$BP = CQ = x \quad (\text{given}) \quad ①$$

$$\angle DBP = \angle DCQ = \alpha + 60 \quad (\text{from ii)}$$

$$\therefore \Delta PBD \equiv \Delta QCD \quad (\text{SAS}) \quad ①$$

(iv) $\angle DPQ = \angle DBC$ (same angle)

$\therefore \angle DPQ = 60^\circ$

$\angle DQC = 180 - \angle DCQ - \angle CDQ$

$= 180 - (\alpha + 60) - \angle BDP$ (1)

$= 180 - (\alpha + 60) - (180 - (\alpha + 60) - 60)$

$= 180 - \alpha - 60 - (60 - \alpha)$

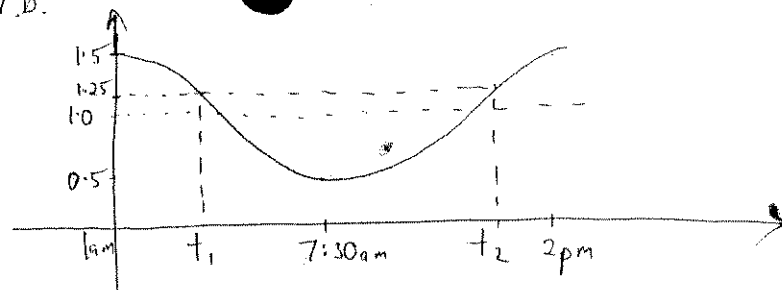
$= 180 - 120$

$= 60^\circ$ (1)

$\therefore \Delta DPQ$ is equilateral (2 $\angle$'s of 60°).

(v) $z = x + y$ (equal sides of equilateral Δ). (1)

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(i) amplitude = 0.5 m (1)

Period = $13 \text{ hrs} \times 60$

$= 780 \text{ minutes.}$

(ii) Let equation of motion be

$x = A \cos(nt) + C$

$T = \frac{2\pi}{n}$

$780 = \frac{2\pi}{n} \therefore n = \frac{2\pi}{780} = \frac{\pi}{390}$

$\therefore x = 0.5 \cos\left(\frac{\pi t}{390}\right) + C$

When $t=0$, $x=1.5$

$\therefore 1.5 = 0.5 \cos 0 + C$

$1.5 = 0.5 + C$

$\therefore C = 1$

$\therefore x = 0.5 \cos\left(\frac{\pi t}{390}\right) + 1$ is the equation (1)
of the motion

Need to find when $t = 0$

$$1.25 = 0.5 \cos\left(\frac{\pi t}{360}\right) + 1$$

$$\cos\left(\frac{\pi t}{360}\right) = 0.5 \quad (1)$$

$$\frac{\pi t}{360} = \frac{\pi}{3}, \frac{5\pi}{3}$$

$\therefore t = 120 \text{ minutes}, 600 \text{ minutes after lan}$

no danger of flooding between

3:10 am and 11:50 am (1)

$$\text{D(a)} \quad f'(x) = x^3 + x^2 + x + 1$$

$$f''(x) = 3x^2 + 2x + 1 \quad (1)$$

$$\text{ii) } f'(x) = 0$$

$$x^3 + x^2 + x + 1 = 0$$

$$x^2(x+1) + 1(x+1)$$

$$(x^2+1)(x+1) = 0$$

$$\text{when } x = -1. \quad (1)$$

$$f''(x) = 3x^2 + 2x + 1$$

$$a = 3, \Delta < 0$$

$\therefore$ positive definite

$\therefore$ curve is always concave up. (1)

So curve must have a minimum turning point at $x = -1$.

$$f(-1) = \frac{1}{4} - \frac{1}{3} + \frac{1}{2} - 1 + C$$

$$= -\frac{7}{12} + C \quad (1)$$

$\therefore$ if $C > \frac{7}{12}$, curve will always be above the x -axis and have no real roots.

$$\text{(b) (i) } z^5 = 1 = \cos 0 + i \sin 0$$

$$z = r(\cos \theta + i \sin \theta)$$

$$\therefore z^5 = r^5(\cos 5\theta + i \sin 5\theta) \quad (1)$$

$$\therefore r^5 = 1 \quad \therefore r = 1$$

$$5\theta = 0 + 2k\pi$$

$$\theta = \frac{2k\pi}{5}$$

$$\therefore z^5 = \cos \frac{2k\pi}{5} + i \sin \frac{2k\pi}{5} \quad \text{for } k=0,1,2,3,4$$

①

When

$$k=0 \quad z_1 = \cos 0 + i \sin 0$$

$$k=1, \quad z_2 = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$$

$$k=2, \quad z_3 = \cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5}$$

$$k=3, \quad z_4 = \cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5} = \bar{z}_3$$

$$k=4, \quad z_5 = \cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5} = \bar{z}_2 \quad \text{①}$$

$$\therefore z^5 - 1 = (z - z_1)(z - z_2)(z - z_3)(z - z_4)(z - z_5)$$

$$z_1 = 1, \quad z_4 = \bar{z}_3, \quad z_5 = \bar{z}_2 \quad \text{①}$$

$$\therefore z^5 - 1 = (z - 1)(z - z_3)(z - \bar{z}_2)(z - z_2)(z - \bar{z}_3) \quad \text{①}$$

$$\begin{aligned} \therefore \frac{z^5 - 1}{z - 1} &= [z^2 - z(\bar{z}_2 + z_2) + z_1 z_2][z^2 - z(z_3 + \bar{z}_3) + z_3 \bar{z}_3] \\ &= [z^2 - z(2\cos \frac{2\pi}{5}) + 1][z^2 - z(2\cos \frac{4\pi}{5}) + 1] \quad \text{①} \\ &= (z^2 - 2z\cos \frac{2\pi}{5} + 1)(z^2 - 2z\cos \frac{4\pi}{5} + 1) \quad \text{①} \end{aligned}$$

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iii) For $z = 1$,

Sum of roots is

$$z_1 + z_2 + z_3 + z_4 + z_5 = 0$$

$$\text{i.e. } z_1 + z_2 + \bar{z}_2 + \bar{z}_3 + z_3 = 0$$

$$1 + 2\cos \frac{2\pi}{5} + 2\cos \frac{4\pi}{5} = 0$$

$$\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2} \quad \text{①}$$

Sum of pairs of roots for $z^5 - 1 = 0$

$$z_1 z_2 + z_1 \bar{z}_2 + z_3 z_1 + z_1 \bar{z}_3 + z_2 \bar{z}_2 + z_2 z_3$$

$$+ z_2 \bar{z}_3 + \bar{z}_2 z_3 + \bar{z}_2 \bar{z}_3 + z_3 \bar{z}_3 = 0$$

$$2\cos \frac{2\pi}{5} + 2\cos \frac{4\pi}{5} + 1 + z_2(2\cos \frac{4\pi}{5}) + \bar{z}_2(2\cos \frac{4\pi}{5}) + 1 = 0 \quad \text{①}$$

$$-1 + 1 + (2\cos \frac{4\pi}{5})(2\cos \frac{2\pi}{5}) + 1 = 0 \quad (\text{from above}),$$

$$\therefore \cos \frac{4\pi}{5} \cos \frac{2\pi}{5} = -\frac{1}{4} \quad \text{①}$$

$\therefore$ The quadratic equation whose roots are $\cos \frac{2\pi}{5}, \cos \frac{4\pi}{5}$ is

$$x^2 - (-\frac{1}{2})x + (-\frac{1}{4}) = 0$$

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